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The Editors
Physical Review E

Dear Editors,

Please consider the manuscript "Relativistic detuning, electron heating and the reach of continuum kinetic methods in resonant dark-photon conversion in a plasma" for publication in Physical Review E.

Particle-in-cell (PIC) simulations by Hook, Huang and Shalaby (Phys. Rev. Lett. 137, 071004) report that nonlinear plasma dynamics saturates the resonant conversion of dark-photon dark matter into plasma oscillations. We revisit their resonant case with a noise-free Hermite–Fourier Vlasov solver, relativistic continuum models (a grid solver and a momentum-space Hermite solver) and an independent relativistic PIC code. The manuscript reports three results. First, relativistic detuning, absent from nonrelativistic models, sets the collapse of the mean-field response and its timing in the continuum models. Second, the electron heating of Hook et al. (re-digitized from their figure) is reproduced to a factor of about two by the converged relativistic PIC, while a Newtonian pusher overestimates the relativistic heating by a factor of about two and a half; the remaining factor between our plateau and their curve is unresolved, and we do not claim that either is wrong. Third, the deterministic continuum models resolve the detuning but give a heating several hundred times smaller than the PIC, and with PIC-like noise they lose positivity before heating develops, which we pose as an open question about the fluctuations that produce the heating. Earlier results are retained in condensed form: verification of the nonrelativistic solver, the truncation instability that limits it, a finite reservoir and finite coherence time, and collisional absorption, now explicitly scoped as statements about a nonrelativistic model. Failed and amended checks, including the new relativistic and PIC-convergence gates that failed, are reported in a table.

All code, records and the scripts that generate every figure, table and quoted number are in the public repository `uwplasma/Dark-SPECTRAX`. The work has not been published or submitted elsewhere.

Sincerely,

Rogerio Jorge
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